

Physical activity interventions for African American women for health and wellness: A scoping review

Journal of Health Psychology

2026, Vol. 31(5) 1789–1804

© The Author(s) 2025

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/13591053251375440

journals.sagepub.com/home/hpq



LaShawn D. Nastvogel¹ 

Abstract

African American women are less likely to be physically active which is one of several lifestyle influential factors affecting this population's health. However, aspects of wellness such as movement, behavioral health, breathing and sleep quality are not being measured but might also contribute to overall health. The purpose of this scoping review is to identify the physical and psychological characteristics of recent physical activity interventions for African American women by documenting intervention characteristics, mapping their results, and acknowledging potential gaps that can be used for future interventions where the emphasis is not only on physical activity but also functional movement and wellness. Eleven articles met the inclusion criteria for this review. Included articles focused on aspects of intrinsic motivation, social support, and goal setting to increase physical activity levels. Group discussions, messaging, and exercise sessions were the most common activities with a focus on providing health and physical activity information.

Keywords

health and wellness, wellbeing, African American women health interventions, functional wellness, physical activity

Introduction

Common chronic conditions such as cardiovascular disease, stroke, renal disease, and diabetes are subject to marked racial and/or gender groups disparities in the United States (Carnethon et al., 2017). Regular, moderate to vigorous physical activity (MVPA) is a cornerstone of physical and mental health promotion and can minimize risk factors such as increased body weight, hypertension, and diabetes (Paluch et al., 2024). Based on data for African American (AA) adults, the prevalence of obesity was 38% and 58% for hypertension in the United States for 2021–2023

(Centers for Disease Control and Prevention, 2023). Further, physical activity (PA) levels differ among racial groups with AA women having lower levels of activity compared to women in other racial groups at all ages, accounting for some health and life expectancy disparities

¹University of Maryland Eastern Shore, Princess Anne, US

Corresponding author:

LaShawn D. Nastvogel, Department of Kinesiology,
University of Maryland Eastern Shore, 1 Backbone Road,
Princess Anne, MD 21853, US.

Email: ldnastvogel@umes.edu

(Centers for Disease Control and Prevention, 2024; Office of Minority Health, 2023). However, interventions to increase PA have not yet fulfilled potential to improve health and wellbeing of AA women.

AA women are willing to engage with interventions focused on health and wellness (Mama et al., 2020). Several limitations contribute to the mediocre effectiveness of existing interventions. In a review by Obi et al. (2023), common barriers to PA faced by AA women include lack of an exercise partner(s), body image perception, haircare, gender norms, family obligations, fear of sexual stereotypes, and environmental barriers. Religiosity, peer pressure with musculoskeletal issues and weight gain as reasons for not engaging in walking as a mode of PA (Obi et al., 2023). Several interventions for AA women have included elements such as education and counseling on eating choices (Brewer et al., 2022) and cultural habits and traditions (Wilcox et al., 2010). Other interventions for AA women have adopted implementing interventions in regularly visited locales such as churches (Collins et al., 2021a); Joseph et al., 2023; Thomas et al., 2017; Whitt-Glover et al., 2017) or hair salons (Palmer et al., 2021).

PA interventions with AA women often use group exercise (Cho et al., 2022) or walking sessions (Pekmezi et al., 2020) and those interventions that have included strength training (Benitez et al., 2024), have included it as a part of the group exercise session or in a supervised one-on-one format. However, few of the interventions with AA women have extended beyond an 18-month period with reported changes in lifestyle-related cardiovascular disease risk factors such as body composition (fat versus fat-free mass), blood pressure, functional wellness or health-related quality of life. So, while AA women may experience modest weight loss or an increase in motivation and PA levels for a given intervention, this same intervention may neglect issues related to sleep quality, completing daily activities without pain or stiffness, and education on how to maintain weight loss. As participants lose weight, their resting energy expenditure, also known as resting metabolic

rate, may also decrease (Hunter et al., 2008). This means that it may become more difficult for participants to maintain or continue losing weight once the intervention concludes.

The largest reductions in cardiovascular disease risk factors are associated with engaging with both aerobic and resistance training activities (Paluch et al., 2024) and therefore, without neglecting cultural components, interventions must consider the incorporation of resistance training exercises as a measurable outcome to assist with maintaining one's metabolic rate, functional capability and capacity (Huang et al., 2022). Maintaining or improving functional capability and capacity may also have a positive effect on musculoskeletal health. In a review of motivational factors of AA women to engage in resistance training, Vasudevan and Ford (2022) found social support such as camaraderie during training; psychological support such as improving body image and mental health; and increased knowledge and awareness of musculoskeletal health helped to encourage and facilitate strength training and weight loss efforts. On the contrary, one pilot intervention for AA women has considered influencing the type of motivation they have towards engaging in PA (Sweeney et al., 2021). By using a challenge-focused group and a rewards-focused group, Sweeney et al. (2021) had over 75% attendance and participants were engaged in the group sessions and self-monitored PA for the entire 6-weeks of the program. When motivation is extrinsic, dependent and/or controlled by external factors such as weight loss or health risks, it may not be considered an enjoyable or pleasant experience (Ryan and Deci, 2000) thereby limiting the intervention's effectiveness and sustainability. Only one such recent study has been identified that included a measure of life satisfaction specifically targeting AA women in a PA intervention (Cho et al., 2022) however, improvements in quality of life or functional wellness have not been explored as an effect of the PA intervention, whether culturally tailored or not.

PA interventions for AA women have focused on health education, weight loss, diet, and PA however, none of them have assessed

functional wellness and little to none of them have surveyed health-related quality of life measures. So, while several interventions for AA women have been implemented, none have measured nor assessed their functional abilities as it relates to activity. Interventions have explored various types of motivation and behavioral change models for engaging in PA however, wellness and/or wellbeing outcomes of the participants are not commonly reported. This may be due to the multifaceted models that have been proposed for functional wellness which include aspects of health and its relationship to the dimensions of wellness (e.g. physical, social, spiritual) (Kiefer, 2008; Simmons, 1989). PA requires bodily movement and if individuals possess acute or chronic pain with or without movement, they may be less likely to be active despite evidence that PA or exercise can improve functional capacity in untrained populations (Huang et al., 2022). As Bland (2017) concluded, function is a process and the amount of positive or negative shifts in function over time are determined by the individual, their environment, lifestyle, and diet.

Functional wellness in this context refers to the interconnectedness of health and wellbeing in such that as one domain improves, another domain will improve as well. The developers of the Functional Wellness Screen™ (Lee Burton and Gray Cook) devised a set of domains to determine the level of musculoskeletal risk involved when an individual chooses to be physically active. These domains include movement and behavioral health, breathing and sleep quality, body composition, nutritional awareness, and injury history. The domains identified with this screen relate to an individual's intention to engage in a PA program and whether precautions should be taken aside from pre-exercise health risk. The model used for functional wellness is dependent on the activity or intervention. The term *functional* has already been used in medicine and recognized as being necessary to create a framework for understanding how an individual may be at risk for disease by focusing on the individual and related factors (Bland, 2017). Functional assessment tools are becoming an

important component of prescription and therapeutic programs because of their individually-centered approach in design and implementation (Bland, 2017). Therefore, it might be reasonable to include such a component in a health and wellness intervention.

Objective

The purpose of this scoping review is to identify what has been done in this area specifically for African American women in the last 5 years by 1) comparing the characteristics of PA interventions; 2) map the results; and 3) identify gaps in the literature that provide direction for future interventions for African American women that target health and wellness. The research questions guiding this scoping review were 1) what were the characteristics of PA interventions for African American women within the past 5 years and 2) what were the results of the PA interventions for African American women within the past 5 years?

Methods

The protocol for this scoping review was not registered but it followed the PRISMA-ScR (Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews) checklist (Peters et al., 2020; Tricco et al., 2018). A reference librarian assisted in identifying potential sources through PubMed and Primo Discovery (EBSCOhost, Medline, Ovid, and ProQuest). Specific key terms used to search both databases included PA, exercise training, resistance training, weight training, and community interventions. The phrase 'for African American women' and 'and African American women' were added to all key terms. Dissertations, review articles, books, newsletter articles, book chapters, magazines, journals were excluded. The publication date was limited to years 2019–2024 to provide the most up to date research on PA interventions for AA women. Eligible sources were published in peer-reviewed, open access journals that were available online and in English. Key terms,

phrases and dates were valuable to capture relevant up to date empirical research. Titles were also used to narrow the search. Titles that included 'African American women' were then critically screened and abstracts were read to determine 1) if a PA intervention was employed, 2) if participants were apparently healthy (did not include special medical conditions, treatments, recovery, or maintenance methods such as osteoarthritis, cancer survivors, cardiovascular disease, asthma, etc.), and 3) if the intervention included only African American women. If meeting the criteria for inclusion could not be determined from the title or abstract, the full text was screened. A sample electronic search that was used for Primo Discovery is available in the supplementary file. An advanced search by title, date and material type was also used to capture as many relevant results as possible.

Eligibility criteria

All articles were independently screened by the author to determine inclusion eligibility. Exclusion criteria included: 1) protocols; 2) rationale or design for an intervention; 3) baseline characteristics of targeted population for a future intervention; 4) simulations; 5) interventions for AA women with special medical conditions such as osteoarthritis, breast cancer, asthma, etc. Eligibility criteria included: 1) publication years 2019–2024 in a peer-reviewed journal; 2) open-access and available online; 3) offered to AA women only; and 4) implemented a PA program using a single group, randomized, nonrandomized, or quasi-experimental interventional design. No limits were placed on participant age. The full texts of the 21 potentially eligible articles were then critically read to confirm inclusion.

Of the 21 articles, 10 were excluded for the following reasons: one did not include only AA women; two were protocols only and the intervention had not occurred yet; one was the rationale for how to set up an effective intervention; one published baseline characteristics of the targeted population but reported the results in another study; one was a simulation prior to

implementation; and four were interventions specifically for AA women with special conditions.

Data items

The characteristics of the studies were summarized and included in Table 1. Specific characteristics extracted from the articles included the purpose of the intervention, the setting, a description about the intervention, what was measured, the findings and whether the intervention was incentivized. The goal of each article was reported first along with where the intervention took place, a description of what the intervention included, what was measured, the findings, and whether it was incentivized. The purpose of the intervention is defined as what the researcher(s) were trying to do (i.e. improve PA, enhance motivation towards PA). The theoretical framework was useful to identify because it was the guide for how the intervention was organized and what activities and instruments were included. The setting is defined as where the intervention activities took place. The description of the intervention includes the design, duration, sample size and mean age range, and all activities that were completed. All data that were collected and measured were also included such as survey items, anthropometrics, and/or accelerometers. How the data were collected was also identified within the intervention description or measurements category. The findings are defined as the results of the intervention such as weight loss or a significant increase in motivation towards PA.

Results

The screening process was completed independently by the author over several days. This poses a significant limitation due to the lack of diverse perspectives. Therefore, the author maintained strict conformity to the inclusion and exclusion criteria and clearly outlined the method of selection. External feedback was received on the screening and selection of sources (Figure 1). Per external feedback,

Table 1. Summary of characteristics.

References	Goal(s)	Intervention setting	Intervention description	Measures	Main findings	Incentivized
Arad et al. (2020) (parent study Arad et al. (2015))	Re-examine previous results of the effectiveness of HIIT on exercise adherence	Hospital	14 weeks 20–40 years old HIIT group ($n=14$) vs control ($n=14$) 24 min for 3 times per week on a cycle ergometer @ 75–90% heart rate reserve	- Adherence - Target heart rate achievement - RPE at each stage	Intervention group - HIIT may be too aggressive - High attrition rate	Yes
Babatunde et al. (2020) (protocol published separately Bevel et al. (2018))	Assess the impact of the SISTAS intervention study on biological markers, plasma C-reactive protein and interleukin-6 Increase PA and healthy dietary habits to reduce breast cancer risk	Church	12 months 39–61 years old 3 months of classes (2-hr per week) and 9-mths of monthly 'booster' sessions (2-hr per month) Intervention group ($n=176$) vs control ($n=161$) Education, cooking classes, and PA sessions	- anthropometrics - BJA - BP - blood samples - (C-reactive protein and interleukin 6 which are inflammatory biomarkers) - psychological and psychosocial variables - sleep quality - diet - exercise Health capability assessments: - push up - curl up - sit and reach test - BP - BMI - Life satisfaction	Both groups - CRP was significantly higher in those with greater obesity, BF, and waist circumference - Intervention group improved diet - Intervention deemed not successful at reducing CRP or IL-6 - PA not reported	Yes
Cho et al. (2022)	Determine the effects of PA intervention on health capabilities and life satisfaction	University campus	12 weeks 32–65 years old $n=35$ Participate in two 50-minute PA sessions per week	- BMI - Leisure PA Survey data - knowledge of PA - psychological variables	Small increases <1 - pushups - curl-ups - sit and reach Improved - BP - BMI - Life satisfaction Intervention group increased PA and frequency Between groups - No difference in knowledge of PA nor beliefs - Significant difference in self-efficacy and self-regulation	No
Collins et al. (2021 a,b)	Increase knowledge and change beliefs about PA Improve social facilitation, self-regulation, self-facilitation to promote increase in PA	Church	6 weeks 35–69 years old Church-based, culturally sensitive intervention ($n=21$) vs control ($n=11$) 4 1-hr sessions (discussions, lectures) Handouts, email reminders on PA once a week	- BMI - Leisure PA Survey data - knowledge of PA - psychological variables	Intervention group increased PA and frequency Between groups - No difference in knowledge of PA nor beliefs - Significant difference in self-efficacy and self-regulation	Yes

(Continued)

Table 1. (Continued)

References	Goal(s)	Intervention setting	Intervention description	Measures	Main findings	Incentivized
Joseph et al. (2021)	Improve PA to national guidelines standard of 150min/wk by walking	Variable/Virtual	16 weeks 50–65 years old ($n = 20$) 14 video and text-based weekly modules focused on promotion of PA knowledge and behaviors Smartphone app, Fitbit PA reminders every 2 weeks	- Anthropometrics - self-reported MVPA - SCT constructs (i.e. self-regulation, confidence) - acceptability and feasibility of app	Increased PA from 20 to 50 minutes per week Increased total weekly energy expenditure (MET) Improved self-regulation and behavioral capability Decreased self-efficacy for PA Decreased use of app as intervention progressed	Yes
Joseph et al. (2023)	Increase PA Reduce cardiometabolic disease risk factors	Variable/Virtual	8-month 24–49 years old culturally tailored smartphone intervention <i>Smart Health wellness</i> ($n = 30$) vs <i>Smart Walk PA</i> ($n = 30$) Smartphone app, Fitbit and Actigraph GT9X link (<i>Smart Walk only</i>) 14 video and text-based weekly modules focused on promotion of PA knowledge and behaviors 3 PA text messages per week (months 1–4)	- self-reported MVPA - cardiometabolic risk markers - anthropometrics - fitness - pulse wave velocity - blood draws - diet	PA group - substantially increased self-reported PA - increased self-efficacy - improved PA regulation Improved fitness, BP, BMI but not statistically significant 96% of participants agreed that text messaging was helpful and motivating App use and Fitbit wear decreased as intervention progressed	Yes
Kitzman et al. (2021)	Improve weight loss and PA	Church	10 months 48.8 ± 11.2 years old 11 churches ($n = 221$) A Health coach(es) at each church and participants received a curriculum binder Evidence-based lifestyle self-monitoring intervention (4 mth. intervention, 6 mth. maintenance) Faith-based ($n = 119$) vs standard ($n = 102$) For faith-based group, 5 additional religious activities were added	- anthropometrics - BP, hemoglobin A1c, fasting glucose, lipoprotein (LDL) and high-density lipoprotein (HDL), and total cholesterol - Process evaluations (dose and fidelity) - self-reported PA - diet	Both groups - lost weight - increased average weekly PA - improved diet - reduced chronic disease risk factors - The faith-based group lost significantly more weight - 20% shifted to a lower BMI category	Yes

(Continued)

Table 1. (Continued)

References	Goal(s)	Intervention setting	Intervention description	Measures	Main findings	Incentivized
Pekmezci et al. (2020)	Increase weekly MVPA	Variable (Home)	12 months 50–69 years old <i>n</i> = 65 Home-based individually tailored vs control Pedometers, print materials/tips sheets, manuals	- 7-Day PAR - 6-min walk test - Minutes of PA via Actigraph GT3X - Time spent strength and flexibility training - psychosocial variables	Both groups - increased self-reported PA - Home-based group - increased PA and time strength and flexibility training more than control - higher goal setting, enjoyment, motivation Control group - lower 6min walk test	Yes
Sweeney et al. (2021)	Increase PA among inactive AA women by tailoring the intervention to their degree of self-motivation, or autonomy	Community center	6 weeks 34–64 years old <i>n</i> = 15 Weekly 90-minute sessions for 6 weeks (feedback, health education, group PA, and behavioral skills training) Challenge-focused group and a rewards-focused group FitBit tracking, accelerometers	- systematic observation - autonomous motivation - BMI - 7-day PA (Actual) - daily steps (FitBit)	Both groups - increased MVPA - increased daily steps	Yes
Sweeney et al. (2024) (published protocol Sweeney et al. (2023))	Assess implementation and formative feedback from a group-based social affiliation PA intervention	South Carolina metropolitan areas	10-week cohort 27–76 years old <i>n</i> = 54 2-hours twice a week Group feedback and problem-solving (15 min), discussion-based health curriculum (30 min), competitive PA session (30 min), group-based behavioral skills training	- Attendance tracking (sign in/out) FitBit - daily steps - PA - Systematic observations - Formative process evaluation (open-ended questions about barriers)	Formative program results (<i>intervention in-progress</i>) - 84.8% weekly attendance - and average of 8.6 out of 10 sessions were attended (with makeup sessions included 97% attendance and 9.5 out of 10 sessions attended) - High levels of reach, dose, and fidelity	No
Zhang and Jemmott (2019)	Increase light PA using app-based group support	Variable/Virtual	3 months 18–35 years old Mobile app-based group (<i>n</i> = 44) vs control (<i>n</i> = 44) 2-hr orientation (PA, self-efficacy) Activity tracking via actigraphy	- BMI - Push-up test 1 min - Daily steps - Minutes of PA via Fitbit - App logins - Coded messages	Both groups - lowered BMI - increased pushups - increased light PA to 90min/day Mobile app > control in PA	Yes

PA: physical activity; BMI: body mass index; MVPA: moderate to vigorous physical activity; PAR: physical activity recall; BIA: bioelectrical impedance analysis; BP: blood pressure; BF: body fat; HILT: high intensity interval training; SCT: social cognitive theory.

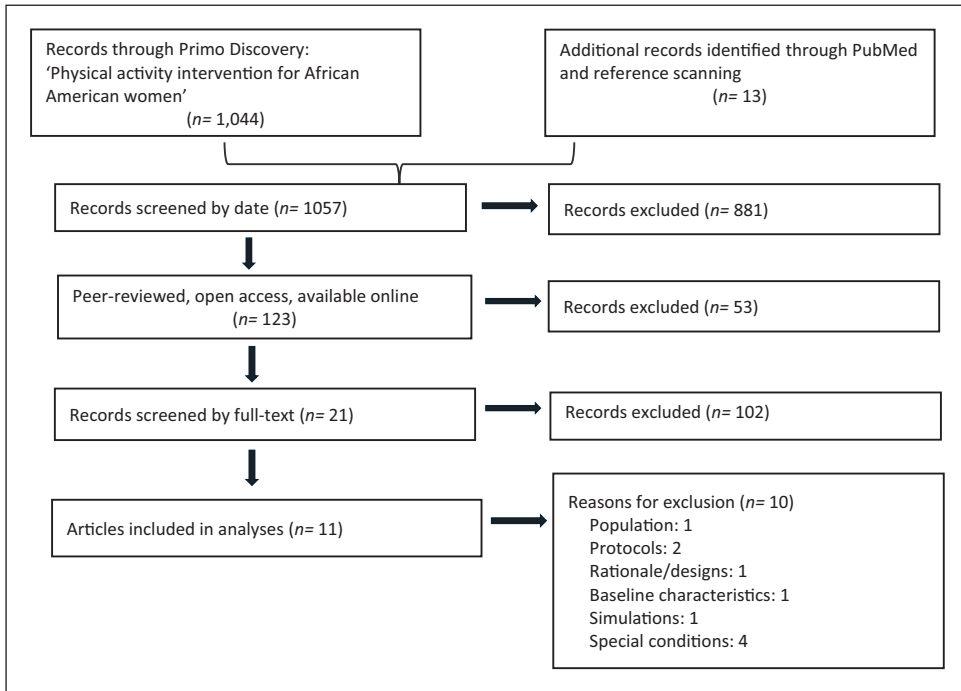


Figure 1. Selection of sources of evidence.

doubtful cases were simply included. Figure 1 depicts the process used for screening the articles that would be included.

Synthesis of results

The characteristics of the study such as their goals, geographic location, structure of PA program, measures, findings, and whether it was incentivized were documented. The content was condensed to key applicable details such as what was done, what was measured, and what was found. Articles were grouped according to similar goals, activities, measures, and/or findings. The full text of each article that met the inclusion criteria was read several times to ensure the charted characteristics were sufficiently detailed. For those articles that published a separate protocol, both the protocol and published findings were used to identify the relevant characteristics. To identify gaps, each article was re-examined for the domains of functional wellness.

Discussion

Intervention goals and theoretical framework

Increasing or improving the amount of time AA women spend being physically active was the purpose of 64% (7/11) of the interventions (Babatunde et al., 2020; Joseph et al., 2021, 2023; Kitzman et al., 2021; Pekmezi et al., 2020; Sweeney et al., 2021; Zhang and Jemmott, 2019). Although Collins et al. (2021a,b) purpose was to increase knowledge and change beliefs about PA, the prediction was that these factors would promote or result in increased PA. Other goals included determining the effect of high-intensity interval training on insulin sensitivity and cardiorespiratory fitness (Arad et al., 2020), the effect of PA on health and life satisfaction (Cho et al., 2022) and assessing feedback from a group-based intervention focused on social affiliation (Sweeney et al., 2024). The group-based intervention focused

on social affiliation is a 5-year ongoing randomized group-cohort trial where researchers are completing formative assessments with each cohort with the goal of improving the intervention for the next cohort.

In terms of the theoretical frameworks, six of the 11 articles created PA interventions through the lens of the social cognitive theory (SCT) (Joseph et al., 2021, 2023; Pekmezi et al., 2020; Sweeney et al., 2021, 2024; Zhang and Jemmott, 2019). The social cognitive theory, developed by Bandura (1986), was used to promote increased participation, engagement, and motivation. Constructs of the social cognitive theory that were directly targeted and measured included behavioral capability, self-efficacy, self-regulation, social support, and outcome expectations ($n=3$). Articles using the SCT incorporated unique multimedia modules for behavioral capability, self-efficacy, and social support; video messages, activity trackers, and discussion boards for self-efficacy, social support, and outcome expectations; and FitBits and print materials for self-regulation. So, while six articles used SCT as a part of the framework, only three of the six measured the constructs of SCT. The three articles that did not measure the constructs of SCT (Sweeney et al., 2021, 2024; Zhang and Jemmott, 2019), integrated one or more of the constructs into the intervention. For example, Sweeney et al. (2024) placed an emphasis on social support by having participants meet as a group weekly and then connect with each other on a mobile app. Outcomes mainly focused on their attendance and delivery of the entire program as intended. Other theoretical frameworks used included the social determination theory (Sweeney et al., 2021), the general interest theory (Sweeney et al., 2021), the PA integrated theory of behavior change (Collins et al., 2021a,b), and the group dynamics theory (Sweeney et al., 2024). Two (Joseph et al., 2021, 2023) reported using Resnicow's cultural tailoring framework by utilizing culturally relevant images, statistics, and staff representation, and one (Babatunde et al., 2020) developed the intervention using a community-based participatory approach by forming Community and Professional Advisory Panels to

determine the intervention activities and materials, measurements, and the name of the study. Three of the 11 articles did not report using a theoretical framework to guide the development of the intervention.

Intervention activities and cultural sensitivity

Group discussions, messaging, and/or exercise sessions were the most common (8/11; 73%) type of intervention activity (Babatunde et al., 2020; Cho et al., 2022; Collins et al., 2021a,b; Joseph et al., 2023; Kitzman et al., 2021; Sweeney et al., 2021, 2024; Zhang and Jemmott, 2019). These were grouped together since an emphasis was placed on the interpersonal or social factors that may positively or negatively influence behaviors. With some of the interventions, participants were grouped randomly whereas other interventions, participants selected their own partner. Three of the interventions included churches as the primary source of recruitment and site of the intervention; therefore, participants were likely familiar with one another (Babatunde et al., 2020; Collins et al., 2021a,b; Kitzman et al., 2021).

Intervention activities that included education sessions focused on topics such as PA and self-efficacy (Kitzman et al., 2021; Pekmezi et al., 2020; Sweeney et al., 2021, 2024; Zhang & Jemmott, 2019) whereas other interventions used and/or supplemented with educational print materials sent via mail or received in person (Babatunde et al., 2020; Collins et al., 2021a,b). Regardless of the delivery method, providing information about health and PA is a common element of interventions as 64% (7/11) of the interventions included this component. Technology-based tracking modalities were used in five of 11 interventions including smart watches, app-based mobile platforms, pedometers and accelerators for tracking and reporting physical activities (Joseph et al., 2021; Joseph et al., 2023; Pekmezi et al., 2020; Sweeney et al., 2021, 2024).

Culturally adapted interventions addressing AA women's values, barriers and challenges

were incorporated in several interventions. For example, Harvey et al. (2016) explored the relationship of engagement in religious practices and PA among AA women ($n=465$) and concluded that social support had a positive significant relationship with religious support. Three interventions used explicit religious components including the pastor and reading bible verses (Kitzman et al., 2021). Other interventions included culturally appropriate images such as AA women engaging in PA or consuming a healthy meal throughout their educational sessions and materials that were used (Collins et al., 2021a,b; Joseph et al., 2021, 2023).

One intervention notably differed from the others. Arad et al. (2020) examined the effects of using high-intensity interval training on an exercise bike (24 minutes, three times per week). It was not clear whether the participants exercised alone or with the other participants when completing their exercise session. No other intervention activities were a part of this intervention.

Objective and subjective measures of the interventions

Over half of the interventions (8/11) included BMI in pre and post measurements although weight loss was not a common goal amongst these interventions. Some Interventions where BMI was an outcome measure, added additional items such as waist circumference (Babatunde et al., 2020; Kitzman et al., 2021), blood pressure (Cho et al., 2022; Kitzman et al., 2021), and glucose and lipids (Joseph et al., 2021, 2023; Kitzman et al., 2021). One intervention targeted inflammatory biomarkers, c-reactive protein and interleukin 6 (Babatunde et al., 2020). Those interventions that tracked PA, whether technology or self-reported, were reported. Watch-based devices (Joseph et al., 2021, 2023; Sweeney et al., 2021; Sweeney et al., 2024; Zhang & Jemmott, 2019) and accelerometers (Joseph et al., 2023; Pekmezi et al., 2020; Sweeney et al., 2021, 2024) were used to monitor daily steps and PA of the participants as post assessment measures. Self-reported measures of PA either included 7-day or 24-hour recall

(Babatunde et al., 2020; Joseph et al., 2021, 2023; Kitzman et al., 2021; Pekmezi et al., 2020). Three interventions incorporated additional fitness measures such as a walk test (Pekmezi et al., 2020), push-up (Cho et al., 2022; Zhang & Jemmott, 2019) and curl-up (Cho et al., 2022) assessment. Arad et al. (2020) measured intervention adherence, achieving a targeted heart rate range and success rate during the intervention, and the average ratings of perceived exertion at each stage of the training session. Arad et al. (2020) focused on metabolic flexibility, insulin sensitivity and cardiorespiratory fitness rather than weight loss.

Many interventions included self-reported measures related to motivation, self-efficacy, self-regulation, and/or goal setting for PA (Babatunde et al., 2020; Collins et al., 2021a,b; Joseph et al., 2021, 2023; Kitzman et al., 2021; Pekmezi et al., 2020). Other measures included life satisfaction (Cho et al., 2022), stress (Kitzman et al., 2021), and diet (Babatunde et al., 2020; Kitzman et al., 2021). Interventions by Joseph et al. (2021, 2023) and Sweeney et al. (2021, 2024) focused on feasibility, acceptability, and process evaluations which provided feedback on how to modify the intervention. For example, Joseph et al. (2021) was a single-arm feasibility trial whereas Joseph et al. (2023) was a pilot trial with a comparison group. Both trials examined the use of an app-based platform, *Smart Walk*, as the primary intervention. Sweeney et al. (2021) was the pilot intervention that informed years 1 and 2 of the 5-year group-cohort design (Sweeney et al., 2024).

Improvements in physical activity and self-efficacy

Most interventions monitored the amount of PA, whether objective or subjective or both. The daily and/or weekly time spent in PA increased in 6/7 interventions addressing PA (Joseph et al., 2021, 2023; Kitzman et al., 2021; Pekmezi et al., 2020; Sweeney et al., 2021; Zhang & Jemmott, 2019). Although not statistically significant, Collins et al. (2021a,b) reported increased PA participation time and frequency despite the

Table 2. Appraisal of sources within domains of functional wellness.

First author	Movement	Behavioral health	Injury history	Breathing quality	Physical activity	Body composition	Sleep quality	Nutrition
Arad	No	No	Yes	Unclear	Yes	No	No	Yes
Babatunde	No	No	Yes	No	No	Yes	No	Yes
Cho	No	Yes	Unclear	No	Unclear	No	No	No
Collins	Unclear	No	Unclear	No	No	No	No	No
Joseph ^a	No	Yes	Yes	No	Yes	No	No	No
Joseph ^a	No	Yes	Yes	No	Yes	No	No	No
Kitzman	No	No	Unclear	No	No	No	No	No
Pekmezi ^a	No	No	Yes	No	Yes	No	No	No
Sweeney	No	Yes	Yes	No	Yes	No	No	No
Sweeney	Unclear	No	Yes	No	Yes	No	No	No
Zhang ^a	No	No	Yes	No	Unclear	No	No	No

^adenotes virtual/home-based intervention.

intervention being as short as 6 weeks. All interventions that focused on self-efficacy, self-regulations, or motivation, reported some degree of improvement in these areas.

Dimensions of functional wellness

While functional wellness was not an outcome of the interventions included in this review, some domains of functional wellness were commonly recognized and some were not. Table 2 displays the domains of the functional wellness screen and whether it was assessed as a part of the intervention. The movement category is a screen to determine the level of musculoskeletal risk in terms of flexibility, mobility, and balance. The behavioral health category uses a survey to determine the ideal state of physical and mental health of the individual. The injury history category is a self-reported survey of the individual’s past musculoskeletal issues. The breathing and sleep quality categories are also survey based and provide feedback related to movement, behavioral health, and physical activity. The PA category assesses the current amount of activity the individual participates in. Body composition is often used interchangeably with body mass index although they are different. Body composition refers to the distribution of fat mass and lean mass with

lean mass including muscle, bone, water, and other non-fat tissues. The nutrition category assesses the amount of knowledge the individual possesses about proper nutrition. The importance of each domain as it relates to health and wellness, not only for the AA population but for all, is outside the scope of this review however, it is clear what has not been addressed in recent PA interventions for AA women. Given the connection between health and wellness, a holistic approach to combatting inactivity may be found in addressing one or more of the domains.

Conclusions

The purpose of this scoping review was to identify the characteristics of PA interventions for AA women that have been conducted in the last 5 years, map their results, and identify gaps. It is important to note that Bland and Sharma (2017) completed a systematic review of PA interventions for AA women that took place between 2009 and 2015 however, the studies were not inclusive of AA women only and the main purpose was to determine if PA increased by assessing specific aspects of the intervention (i.e. duration, frequency, intensity). There was, however, an integrative review of PA interventions that focused only on AA women for the

years 2000–2015 (Jenkins et al., 2017) and therefore this review includes characteristics of more recent interventions.

Six key points can be drawn from this review. First, some but not all interventions for AA women were randomized control trials. While the advantages of randomized control trials (RCT) are well known in interventions, other evaluation methods are important for community engaged work with minoritized communities. Additionally, RCTs have limitations for the control group and generalizability is questionable. Second, most were completed in traditional southern states however, the prevalence of obesity among AA adults in the North and East regions of the U.S. is just as high or higher (Centers for Disease Control and Prevention, 2023). Since walking was the primary mode of PA, weather, such as cold temperatures in the winter, might become an additional barrier for AA adults resulting in the need for indoor accommodation. Third, group-based structures were important for targeting the social support needs of AA women as social support from family, friends, and churches are positively correlated with PA behaviors (Derboghoossian et al., 2024). Namely, partner or small group-based PA has been shown to provide encouragement and motivation for some individuals (Affuso et al., 2022). Fourth, it is crucial to provide access to relevant and culturally appropriate education and/or materials that emphasize the importance of PA for AA women (Joseph et al., 2017). Nine of the 11 interventions included providing information about health and PA whether through messaging, lectures, or by mail. Fifth, since there was no consistent duration for interventions, further research is needed to determine what is effective and sustainable for the AA population. Lastly, despite many of the interventions seeking to improve PA, they were also focused on improving self-efficacy, self-regulation, goal setting, motivations, etc. Although health-related quality of life measures may help identify, understand, and create targeted intervention programs that seek to improve daily living (Bujang et al., 2023; Skevington et al., 2024), an evaluation of functional wellness might also

be valuable for program design. For example, an individual who experiences low back pain may be challenged to increase PA because an increase in PA may aggravate or increase discomfort. Identifying this prior to activity may help inform intervention programming.

If prior studies were used for determining the efficacy of future interventions for AA women, standardization would be difficult as none of the interventions used the same survey instrument; at least 19 different instruments were used among the 11 studies. None of the interventions assessed participants' functional movement which identifies aspects of an individual's movement pattern that may also be a cause for bodily pain or discomfort. Only two of the interventions (Babatunde et al., 2020; Pekmezi et al., 2020) had a maintenance or booster period in which facilitators checked in with the participants, however, neither went beyond 12 months. Therefore, future interventions may also consider adding activities that can be maintained and tracked by the participants long-term to help ensure participants have an opportunity to continue positive lifestyle changes.

Implications of the findings

Many PA interventions for AA women have been completed in the southern parts of the United States within the last five years. Dominant characteristics of the interventions in this scoping review included group-based activities (i.e. in-person discussions, group chat via app-based platform) and educational sessions on PA and health. It appears the focus of recent interventions is on time spent being physically active or the motivation to be physically active to improve cardiovascular risk factors such as body mass index (BMI), blood pressure, or glucose whereby neglecting overall wellness. Overall wellness, or quality of life was assessed in two of the interventions, however, the intervention activities only targeted aspects of PA whether exercise or education based. In a review of community-based culturally tailored programs for AA adults, Singh et al. (2023) reported 67.6% ($n=50/74$) of the interventions since 2000 assessed physiological outcomes whereas only 17.6% ($n=13/74$)

assessed quality of life and 6.8% ($n=5/74$) assessed functional performance. Several interventions were successful in improving PA, but it is not clear if many of these activities were developed and included for sustainability purposes, meaning the ability to continue meeting as a group, engaging in group chats via the app-based platform, or exercising in the group setting once the intervention program period concluded. Being that quality-of-life (QoL) assessments may be an indication of an individual's condition, aside from clinical evidence, an aspect of each QoL domain may be worth considering when planning interventions (Bujang et al., 2023). Recent evidence suggested the environmental, social, and spiritual domains of health related QoL are more valuable and better predict women's health (Skevington et al. 2024). To add, global evidence also reported women have poor sleep quality, less energy, and more fatigue than men (Skevington et al., 2024). Therefore, since just two of the interventions in this scoping review assessed QoL and three of the interventions had either a spiritual component (Kitzman et al., 2021) or were delivered in or with a church (Babatunde et al., 2020; Collins et al., 2021a,b), future work will be directed on identifying characteristics of church-based interventions for AA women and how it may influence psychological processes towards health-related lifestyle factors.

Limitations

This scoping review has some limitations. First to make this review contemporary for future intervention planning, the search was limited to the last 5 years, which significantly decreased the number of available articles that included AA women only. Further, to be included in this scoping review, full text needed to be open access and written in English. Interventions that were implemented outside of the United States were not included. There may be additional studies that meet the inclusion criteria, however, only Primo Discovery, a university-based platform, and PubMed were the only databases that were available at the university.

Recommendations for future interventions

Future interventions for AA women could offer PA options in multiple settings within the community such as churches, recreational centers or fields, or community health centers. For example, alternating PA settings in the intervention to minimize monotony and create awareness for the possibility of PA in non-traditional settings conducive and appropriate for AA women. In addition, being able to vary settings while still implementing the intervention may lend itself to sustainability. Several interventions were either virtual, home-based, or required self-directed PA. The premise of encouraging self-directed PA throughout the interventions had to do with long-term motivation and adherence however, the support structure with virtual, home-based, self-guided PA is variable, if at all, thereby neglecting the social aspect of the social cognitive theory which health interventions are often grounded in. Educational sessions and print materials made up a large portion of the interventions with all but two providing some form of health and PA education. Future interventions may continue providing knowledge through educational sessions and print materials that are culturally appropriate but, could consider utilizing active teaching methods or breaking the session into smaller increments, as opposed to only lectures and discussions. Only one intervention incorporated such a method and had over 80% attendance for 10 weeks (Sweeney et al., 2024). Finally, since there was a lack of consistent attention to the domains of functional wellness prior to individuals engaging in PA, future interventions may gain valuable insight on the benefits of comprehensive programming for AA women by focusing on both health and wellness.

Acknowledgements

The author would like to thank Dr. Dennis Fortenberry, MD, MS for his continuous editing, feedback, and mentorship.

ORCID iD

LaShawn D. Nastvogel  <https://orcid.org/0009-0002-1776-4763>

Ethical considerations

Ethics approval was not required for this review article as no data were collected from participants.

Consent to participate

Consent to participate is not applicable to this review article as no data were collected from participants.

Consent for publication

Consent for publication is not applicable to this review article as no identifiable participant data are included.

Funding

The author received no financial support for the research, authorship, and/or publication of this article.

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Data availability statement

The datasets generated during the current study are available from the corresponding author on reasonable request.

Supplemental material

Supplemental material for this article is available online.

References

- Affuso O, Kinsey AW, Whitt-Glover MC, et al. (2022). Social environments and physical activity among black women. *American Journal of Health Promotion* 36(8): 1275–1283.
- Arad AD, Albu JB and DiMenna FJ (2020) Feasibility of a progressive protocol of high-intensity interval training for overweight/obese, sedentary African American women: a retrospective analysis. *BMC Sports Science, Medicine & Rehabilitation* 12(1): 59–59.
- Arad AD, DiMenna FJ, Arad AD, et al. (2015) High-intensity interval training without weight loss improves exercise but not basal or insulin-induced metabolism in overweight/obese African American women. *Journal of Applied Physiology (Bethesda, MD : 1985)* 119(4): 352–362.
- Babatunde O, Adams SA, Truman S, et al. (2020) The impact of a randomized dietary and lifestyle intervention on chronic inflammation among obese African American women. *Women Health* 60(7): 792–805.
- Bandura A (1986) *Social foundations of thought and action: A social cognitive theory*. Englewood Cliffs, NJ: Prentice Hall.
- Benitez TJ, Brown N, Marcus B, et al. (2024) Promotion of muscle-strengthening activity among Latina and Black/African American women: A review of literature. *American Journal of Lifestyle Medicine*. Epub ahead of print 17 April 2024. DOI: 10.1177/15598276241246734.
- Bevel M, Babatunde OA, Heiney SP, et al. (2018) Sistas Inspiring Sistas Through Activity and Support (SISTAS): Study Design and Demographics of Participants. *Ethnicity & disease* 28(2): 75–84.
- Bland J (2017) Defining *function* in the functional medicine model. *Integrative Medicine (Encinitas, Calif.)* 16(1): 22–25.
- Bland V and Sharma M (2017) Physical activity interventions in African American women: A systematic review. *Health Promotion Perspectives* 7(2): 52–59.
- Brewer LC, Bowie J, Slusser JP, et al. (2022) Religiosity/spirituality and cardiovascular health: The American Heart Association life's simple 7 in African Americans of the Jackson Heart Study. *Journal of American Heart Association* 11(17): 1–33.
- Bujang MA, Lai WH, Ratnasingham S, et al. (2023) Development of a Quality-of-Life Instrument to Measure Current Health Outcomes: Health-Related Quality of Life with Six Domains (HRQ-6D). *Journal of Clinical Medicine* 12(8): 2816–.
- Carnethon MR, Pu J, Howard G, et al. (2017) Cardiovascular health in African Americans: A scientific statement from the American Heart Association. *Circulation* 136(21): e393–e423.
- Centers for Disease Control and Prevention (2023) *Adult Obesity Prevalence Maps*. U.S. Department of Health and Human Services. Available at: www.cdc.gov/obesity (accessed 11 June 2024).
- Centers for Disease Control and Prevention (2024) *Physical Activity Basics*. Available at: <https://>

- www.cdc.gov/physical-activity-basics/guidelines/index.html (accessed 11 June 2024).
- Cho D and Kim SK (2022) Effects of exercise training on health capabilities and life satisfaction among overweight and obese African American women. *African Journal of Reproductive Health* 26(3): 54–62.
- Collins ES, Buchholz SW, Cranford J, et al. (2021a) A church-based culturally sensitive physical activity intervention in African American women. *Western Journal of Nursing Research* 43(6): 563–571.
- Collins KA, Fos LB, Ross LM, et al. (2021b) Aerobic, resistance, and combination training on health-related quality of life: The STRRIDE-AT/RT Randomized Trial. *Frontiers in Sports and Active Living* 2: 1–10.
- Derboghoossian G, Foust JB and Hayman LL (2024). Social and religious networks with respect to the health behaviors of African American Women: A systematic review. *Journal of Religion and Health* 64(1): 245–271.
- Harvey IS, Story CR, Knutson D, et al. (2016) Exploring the relationship of religiosity, religious support, and social support among African American women in a physical activity intervention program. *Journal of Religion and Health* 55(2): 495–509.
- Huang J, Zhong M and Wang J. (2022) Effects of exercise-based interventions on functional movement capability in untrained populations: A systematic review and meta-analysis. *International Journal of Environmental Research and Public Health* 19: 9353.
- Hunter GR, Byrne NM, Sirikul B, et al. (2008) Resistance training conserves fat-free mass and resting energy expenditure following weight loss. *Obesity (Silver Spring, Md)* 16(5): 1045–1051.
- Jenkins F, Jenkins C, Gregoski MJ, et al. (2017) Interventions promoting physical activity in African American women: An integrative review. *Journal of Cardiovascular Nursing* 32(1): 22–29.
- Joseph RP, Ainsworth BE, Hollingshead K, et al. (2021) Results of a culturally tailored smartphone-delivered physical activity intervention among midlife African American women: Feasibility trial. *JMIR mHealth and uHealth* 9(4): e27383.
- Joseph RP, Keller C, Affuso O, et al. (2017). Designing culturally relevant physical activity programs for African American women: A framework for intervention development. *Journal of Racial Ethnicity & Health Disparities* 4(3): 397–409.
- Joseph RP, Todd M, Ainsworth BE, et al. (2023) Smart Walk: A culturally tailored smartphone-delivered physical activity intervention for cardiometabolic risk reduction among African American women. *International Journal of Environmental Research and Public Health* 20(2): 1000.
- Kiefer RA (2008) An integrative review of the concept of well-being. *Holistic Nursing Practice* 22(5): 244–252.
- Kitzman H, Mamun A, Dodgen L, et al. (2021) Better me within randomized trial: Faith-based diabetes prevention program for weight loss in African American women. *American Journal of Health Promotion* 35(2): 202–213.
- Mama SK, Bhuiyan N, Chaoul A, et al. (2020) Feasibility and acceptability of a faith-based mind-body intervention among African American adults. *Translational Behavioral Medicine* 10(4): 928–937.
- Office of Minority Health (2023) Obesity and African Americans. Available at: www.minorityhealth.hhs.gov (accessed 16 October 2023).
- Obi OC, Nnonyelu AC, Onobrakpeya A, et al. (2023) Benefits and barriers to physical activity among African women: A systematic review. *Sports Medicine and Health Science* 5: 59–66.
- Palmer KN, Rivers PS, Melton FL, et al. (2021) Health promotion interventions for African Americans delivered in U.S. barbershops and hair salons – a systematic review. *BMC Public Health* 21(1553): 1–21.
- Paluch AE, Boyer WR, Franklin BA, et al. (2024) Resistance exercise training in individuals with and without cardiovascular disease 2023 update: A scientific statement from the American Heart Association.
- Pekmezi D, Ainsworth MC, Desmond R, et al. (2020) Physical activity maintenance following home-based individually-tailored print interventions for African American women. *Health Promotion Practice* 21(2): 268–276.
- Peters MDJ, Marnie C, Tricco AC, et al. (2020) Updated methodological guidance for the conduct of scoping reviews. *JBI evidence synthesis* 18(10): 2119–2126.
- Ryan RM and Deci EL (2000) Self-determination theory and the facilitation of intrinsic moti-

- vation, social development, and well-being. *American Psychologist* 55(1): 68–78.
- Simmons SJ (1989) Health: A concept analysis. *International Journal of Nursing Studies* 26(2): 155–161.
- Singh H, Fulton IVJ, Mirzazada S, et al. (2023) Community-based culturally tailored education programs for black communities with cardiovascular disease, diabetes, hypertension, and stroke: Systematic Review Findings. *Journal of Racial and Ethnic Health Disparities* 10(6): 2986–3006.
- Skevington SM, Schick-Makaroff K, Rowland C, et al. (2024) Women's environmental quality of life is key to their overall quality of life and health: Global evidence from the WHOQOL-100. *PLoS One* 19(10): e0310445.
- Sweeney AM, Wilson DK, Zarrett N, et al. (2021) The feasibility and acceptability of the Developing Real Incentives and Volition for Exercise (DRIVE) Program: A pilot study for promoting physical activity in African American Women. *Health Promotion Practice* 22(6): 840–849.
- Sweeney AM, Wilson DK, Zarrett N, et al. (2023) An overview of the Together Everyone Achieves More Physical Activity (TEAM-PA) trial to increase physical activity among African American women. *Contemporary Clinical Trials* 129: 107207.
- Sweeney AM, Wilson DK, Zarrett N., et al. (2024) Using formative process evaluation to improve program implementation and accessibility of competitive group-based physical activity in the TEAM-PA trial. *The International Journal of Behavioral Nutrition and Physical Activity* 21(1): 88–111.
- Thomas S, Yingling L, Adu-Brimpong V, et al. (2017) Mobile health technology can objectively capture physical activity (PA) targets among African American women within resource limited communities-The Washington, D.C. Cardiovascular Health and Needs Assessment. *Journal of Racial and Ethnic Health Disparities* 4(5): 876–883.
- Tricco AC, Lillie E, Zarin W, et al. (2018) PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and explanation. *Annals of Internal Medicine* 169(7): 467–473.
- Vasudenvan A and Ford E (2022) Motivational factors and barriers towards initiating and maintaining strength training in women: A systematic review and meta-synthesis. *Prevention Science* 23: 674–695.
- Whitt-Glover MC, Goldmon MV, Gizlice Z, et al. (2017) Learning and Developing Individual Exercise Skills (L.A.D.I.E.S.) for a Better Life: A church-based physical activity intervention—Baseline participant characteristics. *Ethnicity & Disease* 27(3): 257–264.
- Wilcox S, Laken M, Parrott AW, et al. (2010) The Faith, Activity, and Nutrition (FAN) Program: Design of a participatory research intervention to increase physical activity and improve dietary habits in African American churches. *Contemporary Clinical Trials* 31(4): 323–335.
- Zhang J and Jemmott JB III (2019). Mobile app-based small-group physical activity intervention for young African American women: A pilot randomized controlled trial. *Prevention Science* 20(6): 863–872.

Copyright of Journal of Health Psychology is the property of Sage Publications Inc. and its content may not be copied or emailed to multiple sites without the copyright holder's express written permission. Additionally, content may not be used with any artificial intelligence tools or machine learning technologies. However, users may print, download, or email articles for individual use.